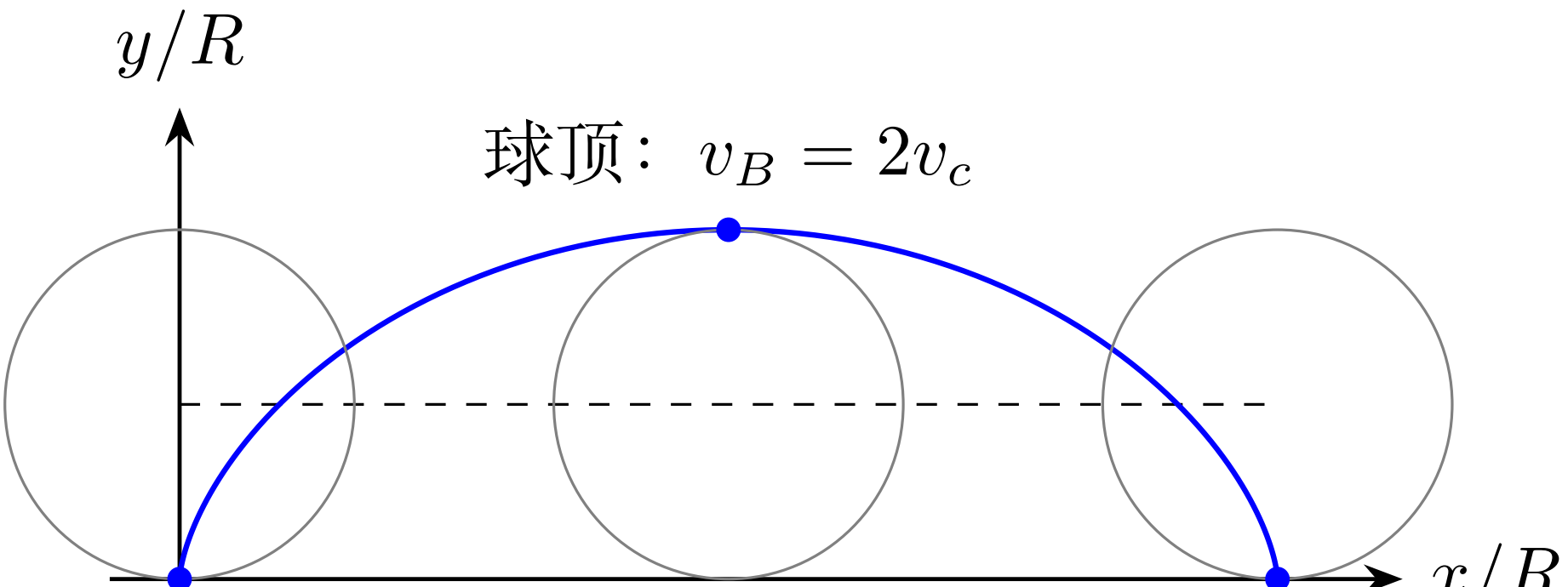

$$x_B/R = \theta - \sin\theta, \quad y_B/R = 1 - \cos\theta$$